



PHASE I · PYTHON FOR AI RESEARCH · CLASS 01

Can you teach a computer to keep score?

Your first Python experiment · student × AI · predict → test → explain

Bo Shang · Merit Point Academy · 60 minutes

No installation needed. Have a keyboard and your curiosity ready.



TODAY'S MISSION

Leave with something you can explain

1 • Predict

Say what a few lines will do before pressing Run.

2 • Experiment

Change one value and explain the new result.

3 • Keep evidence

Save code, output, one limitation and a decision you made after AI helped.

Today: 0–8 warm-up · 8–22 first code · 22–35 data · 35–48 experiment · 48–57 save · 57–60 exit ticket.



CHAT VOTE · 4–8 MIN

Quick check · tap an answer

A bot scored 3 goals in 5 shots. Another scored 4 in 10. Which had the higher success rate?

Bot A      3 goals / 5 shots

Bot B           4 goals / 10 shots

A The first bot

B The second bot

C They are equal



FROM QUESTION TO EVIDENCE

Code is a way to check your idea



01

Ask

How often does the shot go in?



02

Observe

One attempt → one 0 or 1.



03

Test

Count goals and all attempts.



04

Explain

Describe this sample and its limits.

Student: choose a question and make a prediction.

Question → data → test → evidence → a better question.

You choose the question and check the conclusion. AI can help you build a test.



MISSION 1 · 10-14 MIN

Make Python speak for you

Predict the two output lines. Run once. Change only the researcher name, then run again.

```
1 researcher = "Alex"
2 question = "How often did the shot go in?"
3 print("Hello,", researcher)
4 print(question)
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



NAME → VALUE · 14-16 MIN

A variable keeps a value for later

goals is a name; 3 is its value. = assigns a value. Predict the output, then change only goals to 4 and run again.

```
1 goals = 3
2 shots = 5
3 print(goals)
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



PREDICT WITHOUT RUNNING

Quick check · tap an answer

What does `print("3" + "2")` display?

- ☐ A 5
- ☐ B 32
- ☐ C An error



MISSION 2 · 18-22 MIN

Change one thing, check one result

Run the starter. Predict the rate for 4 goals in 5 shots, then change only goals.

```
1 goals = 3
2 shots = 5
3 success_rate = goals / shots * 100
4 print("Goals:", goals)
5 print("Success rate:", success_rate, "%")
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



KEEP THE ATTEMPTS · 22-25 MIN

A list remembers more than the total

1 = goal · 0 = miss · each position is one attempt. Predict both outputs: sum counts goals; len counts attempts.

```
1 shots = [1, 0, 1, 1, 0]
2 print(sum(shots))
3 print(len(shots))
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



MISSION 3 · 25–30 MIN

Add one more attempt

Predict the result if the next shot is a goal. Add one 1 at the end of the list, run, and explain the change.

```
1 # Synthetic practice data: 1 = goal, 0 = miss
2 shots = [1, 0, 1, 1, 0]
3 goals = sum(shots)
4 attempts = len(shots)
5 print("Goals:", goals)
6 print("Attempts:", attempts)
7 print("Success rate:", goals / attempts * 100, "%")
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



BUG DETECTIVE · 30-35 MIN

One letter is hiding the answer

Run this deliberately broken example. Read the final error line and compare the names. Fix only the mistaken variable name.

```
1 goals = 3
2 shots = 5
3 # One variable name below is wrong. Fix only that name.
4 print(goal / shots * 100)
```



Run



Explain each line



Explain



Debug



Improve



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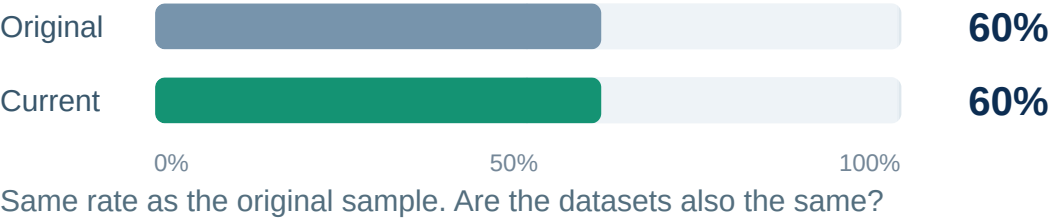
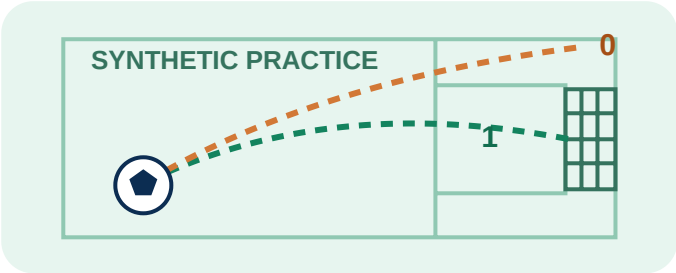
Python loads on first Run

► Run the code to see the output here.



PARTNER CHALLENGE · 35-43 MIN

Change the data. Watch the evidence.



Your attempts

1 = goal · 0 = miss



3 goals 5 attempts 60% success

Predict the rate after one extra goal:

Your %

Test +1 goal

Predict first. Then test and explain what changed.

After your first test, click a shot to change it.

Ask AI: “I predict [your prediction]. Suggest one test that could challenge it. Wait for my result.”
Then explain what changed, and download the Notebook for **this** sample.



WHAT CAN WE CLAIM?

Quick check · tap an answer

Our imaginary player made 4 of 6 shots. Which claim is supported?

- A** This player is always better than everyone else
- B** The success rate in this six-shot practice sample is about 67%
- C** Python proved our training method works



YOUR SMALL RESEARCH NOTE · 46–48 MIN

Question + code + result + limitation

Your checklist

- A question about your practice data
- A list of at least five 0/1 attempts
- The count and success rate
- One change and its effect
- One thing this result cannot prove
- One AI suggestion you tested, changed or rejected

Choose your story

Soccer shots · basketball free throws · a game character's attempts
Keep the data **synthetic** and say so. A new story needs a clear meaning for 0 and 1.



PUT IT TOGETHER

Your first explainable Python experiment

Run the starter, replace the practice data, and add a comment describing your change.

```
1 # Synthetic practice data, not a real player study
2 question = "How often did each practice shot go in?"
3 shots = [1, 0, 1, 1, 0]
4 print(question)
5 print("Goals:", sum(shots))
6 print("Attempts:", len(shots))
7 print("Success rate:", sum(shots) / len(shots) * 100, "%")
8 print("Limitation: this is a tiny synthetic sample.")
```



Run



Explain each line



Explain



Debug



Improve



Mira

Python loads on first Run

► Run the code to see the output here.



SAVE YOUR WORK · 52–57 MIN

Take your Notebook with you

1. Return to [the interactive shot slide](#) and choose **Download Notebook** for your displayed sample.
2. Keep the `.ipynb` file on your computer. Add any further code edits you made in the Python lab.
3. In your [research record](#), save one AI suggestion, your test and your decision; download the record.
4. After class, with help if needed, open [Kaggle Notebooks](#) and import the file.
5. Run it from top to bottom, save a version, and copy your Notebook link.

Your first run already works in the browser. If account setup takes time, keep the downloaded file and finish the Kaggle step with a parent after class.

Pause screen sharing before entering an email, password, or account details.



EXIT TICKET · 57–60 MIN

Explain one line. Predict one change.

Explain

What does `len(shots)` measure?

Predict

What happens when a miss is appended?

Reflect

What can this tiny sample not prove?

Name one decision you made, and one AI suggestion you checked. Submit your saved Notebook link through [course assignments](#) after class.



OPTIONAL RESOURCE · STUDENT × AI

You are the researcher. AI is your collaborator.

Your responsibility

Choose the question. Predict. Decide what counts as a fair test. Check the code and results. Explain the limits.

Ask AI to help

Explain one idea. Suggest an alternative. Implement one small change. Propose a failure case. Help clarify writing after evidence exists.

Keep a suggestion you rejected or corrected. Your judgment is part of the work.

[Open copyable prompts for all 15 lessons ↗](#)



What changes when a computer filters an image?

Predict what will happen to the ball and the field markings. Change one filter, then compare.

Pixel Lab

Simple filters · no trained AI

Filter

Black & white threshold

Brightness cutoff

150

Baseline

Reset

BEFORE

Synthetic soccer scene

AFTER

Threshold · 150

Bright pixels become white; darker pixels become black. Can field lines be mistaken for the ball?

Mean RGB change **79.8** / 255

9.6% white pixels · not a ball detection score

[Larger image lab + AI prompt ↗](#) · [Keeper game ↗](#) · [Camera shake lab ↗](#) · [Five project routes ↗](#)



OPTIONAL RESOURCE · STUDENTS & FAMILIES

What might this interest become?

Games → algorithms and software

Tracking → computer vision, signals and robotics

Movement → mechanics and biomedical engineering

Fair comparisons → statistics and data science

[Open the interactive major-exploration map ↗](#)

Our destination: a reproducible project and a student-written K–12 poster abstract draft for ISEC. Acceptance is decided by conference review.

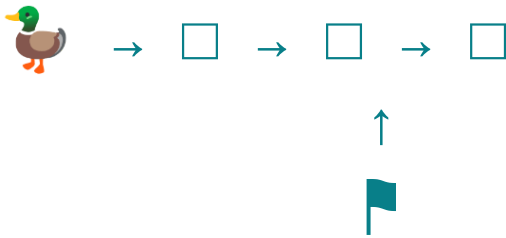


Make the code visible

Write Python, predict a path, then inspect the actual lines, values and moves.

Python in Motion

Sequence → assignment → numbers and text → lists → conditions → loops → functions



Predict. Run real Python. Inspect the executed line and variable values. Change one input or wall, then explain.

[Open all 16 visual concepts ↗](#) · [Keep your experiment record ↗](#)

Optional after the core lesson. Our course follows Tianyi Chai's [Python for AI Research](#) sequence; these visual exercises are original course activities.